

Numerical Methods for Fluid Flow

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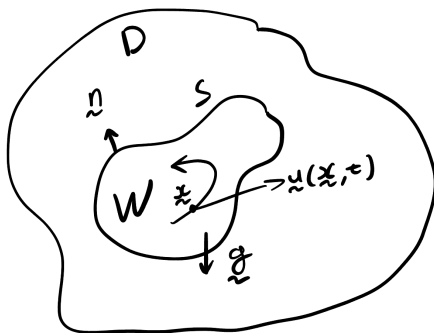
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Overview

1 Background: Fluid Mechanics

2 Computation

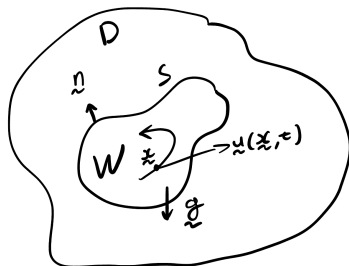
Introduction



Fluid particles flowing in a region D .

- density: $\rho(\vec{x}, t)$
- velocity: $\vec{u}(\vec{x}, t)$
- pressure: $p(\vec{x}, t)$

Conservation of Mass



$$\frac{d}{dt} m(W, t) = \frac{d}{dt} \int_W \rho(\vec{x}, t) dV$$

$$\boxed{\frac{d}{dt} \int_W \rho dV = - \int_S \rho \vec{u} \cdot \vec{n} dA}$$

Law of Conservation of Mass

Conservation of Mass: Continuity Equation

Divergence Theorem: $-\int_S (\rho \vec{u} \cdot \vec{n}) dA = -\int_W \nabla \cdot (\rho \vec{u}) dV$

Leibniz integral rule: $\frac{d}{dt} \int_W \rho dV = \int_V \frac{\partial \rho}{\partial t} dV$

$$\int_W \frac{\partial \rho}{\partial t} dV = -\int_W \nabla \cdot (\rho \vec{u}) dV \Rightarrow \int_W \underbrace{\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u})}_{\text{for all } W} dV = 0$$

$$\Rightarrow \boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) = 0}$$

Continuity Equation

When ρ is constant: $\boxed{\nabla \cdot \vec{u} = 0}$ (Incompressibility)

Material Derivative, Stress Tensor

Material Derivative:

$$\frac{D\vec{u}}{Dt} = \left(\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \right) \vec{u}$$

Main Idea: takes into account the fact that the fluid is moving and that the positions of fluid particles change with time.

Stress Tensor:

$$\sigma = -p\mathbf{I} + \mu \left(\nabla \vec{u} + \nabla^T \vec{u} \right)$$

$$\int_S \sigma \cdot \vec{n} dS = \int_V \nabla \cdot \sigma dV$$

$$\nabla \cdot \sigma = -\nabla p + \mu \nabla^2 \vec{u}$$

Main Idea: describes all internal forces (stresses), Force/Area

Momentum Equation

$$\Delta V$$

$$m = \rho \Delta V$$

$$F_{\text{gravity}} + F_{\text{stress}} = m \vec{a}$$

$$m \vec{g} + (\nabla \cdot \sigma) \Delta V = m \frac{D\vec{u}}{Dt}$$

$$\Rightarrow \rho \Delta V \vec{g} + (\nabla \cdot \sigma) \Delta V = \rho \Delta V \frac{D\vec{u}}{Dt}$$

$$\boxed{\rho \frac{D\vec{u}}{Dt} = \nabla \cdot \sigma + \rho \vec{g}}$$

Momentum Equation

Momentum Equation: $\rho \left(\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} \right) = -\nabla p + \mu \nabla^2 \vec{u}$

$$\vec{u} = (u, v)$$

Navier-Stokes Equations

$$\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = -\partial_x p + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

$$\rho \left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) = -\partial_y p + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right)$$

Assumption 1: Steady Flow ($\frac{\partial}{\partial t} = 0$)

$$\rho \left(\cancel{\frac{\partial u}{\partial t}} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = -\partial_x p + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

$$\rho \left(\cancel{\frac{\partial v}{\partial t}} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) = -\partial_y p + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right)$$

Assumption 2: $p = p(y)$, $v = 0$

$$\rho \left(\cancel{\frac{\partial u}{\partial t}} + u \frac{\partial u}{\partial x} + v \cancel{\frac{\partial u}{\partial y}} \right) = -\cancel{\partial_x p} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

$$\rho \left(\cancel{\frac{\partial v}{\partial t}} + u \cancel{\frac{\partial v}{\partial x}} + v \cancel{\frac{\partial v}{\partial y}} \right) = -\partial_y p + \mu \left(\cancel{\frac{\partial^2 v}{\partial x^2}} + \cancel{\frac{\partial^2 v}{\partial y^2}} \right)$$

Assumption 3: $\nabla \cdot \vec{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$ (Incompressibility)

$$\rho \left(\cancel{\frac{\partial u}{\partial t}} + u \cancel{\frac{\partial u}{\partial x}} + v \cancel{\frac{\partial u}{\partial y}} \right) = -\cancel{\partial_x p} + \mu \left(\cancel{\frac{\partial^2 u}{\partial x^2}} + \boxed{\frac{\partial^2 u}{\partial y^2}} \right)$$

$$\rho \left(\cancel{\frac{\partial v}{\partial t}} + u \cancel{\frac{\partial v}{\partial x}} + v \cancel{\frac{\partial v}{\partial y}} \right) = \boxed{-\partial_y p} + \mu \left(\cancel{\frac{\partial^2 v}{\partial x^2}} + \cancel{\frac{\partial^2 v}{\partial y^2}} \right)$$

$$\boxed{\frac{\partial^2 u}{\partial y^2} = 0}$$

Side note: If gravity considered: $\frac{\partial p}{\partial y} = -\rho g \Rightarrow p = -\rho g y + \text{constant}$

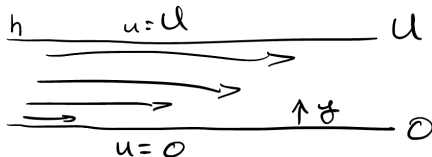
Couette Flow

No-slip Condition: ($u = 0$ on $y = 0$), ($u = U$ on $y = h$)

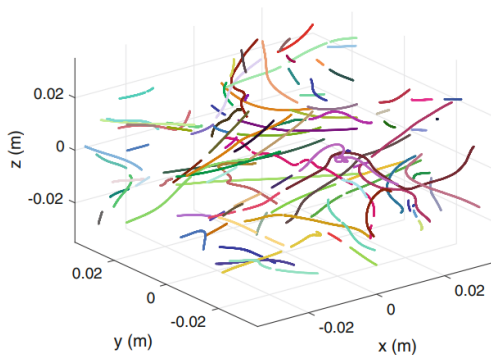
$$\frac{\partial^2 u}{\partial y^2} = 0 \Rightarrow u = Ay + B$$

$$U = Ah \Rightarrow A = \frac{U}{h} \Rightarrow \boxed{u(y) = \frac{U}{h}y}$$

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 & U/h \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



Computational Problem setup



Particle Tracking Velocimetry Data

$$\vec{x}'(t) = \vec{u}(\vec{x}) = A\vec{x}$$

$$\vec{x}(0) = \vec{x}_0$$

$$\vec{x}'(t) = A\vec{x} \Rightarrow \vec{x}(t) = e^{At}\vec{x}_0$$

Estimating A with A_θ

$$A_\theta = \begin{bmatrix} \theta_1 & \theta_2 \\ \theta_3 & \theta_4 \end{bmatrix}$$

We observe $X_{\text{experimental}} = \text{ode45}(A_{\text{groundtruth}}\vec{x}) + N(10^{-2}, 1)$

We estimate A_θ using observed $X_{\text{experimental}}$ values

Constructing a loss function analytically

$$x'(t) = A\vec{x} \Rightarrow \vec{x}(t) = e^{At}\vec{x}_0$$

$$e^{At} = 1 + At + \frac{A^2 t^2}{2!} + \frac{A^3 t^3}{3!} + \dots$$

$$J(\theta) = \frac{1}{2} \int_0^T \|\vec{x}(t) - \vec{x}_\theta(t)\|^2 dt$$

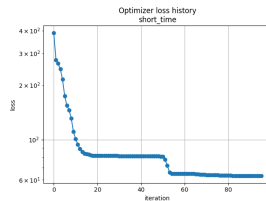
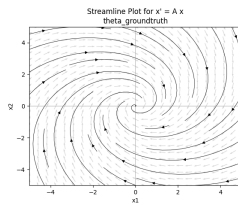
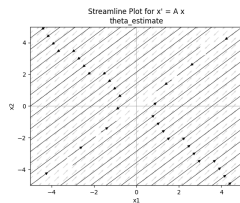
$$J(\theta) = \frac{1}{2} \int_0^t \underbrace{\|((A - A_\theta)t + \dots)\vec{x}_0\|^2}_{\approx t^2 \|(A - A_\theta)\vec{x}_0\|^2} dt$$

$$e^{At} \approx I + At \Rightarrow J(\theta) \approx \frac{1}{2} \|(A - A_\theta)\vec{x}_0\|^2 \int_0^T t^2 dt$$

Short-time

$$\theta_{\text{groundtruth}} = [-0.5, 1, -1, -0.5], \theta_0^* = [0.5, -1, 1, 0.5]$$

```
t_vals = linspace(0, 5, 100)
```



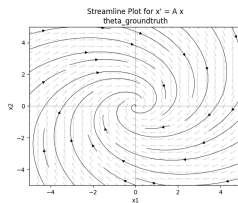
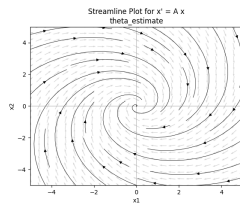
```
final loss = 63.087
```

```
final estimate = [ 179.85, -180.38, 180.32, -180.85]
```


Short-time

$$\theta_{\text{groundtruth}} = [-0.5, 1, -1, -0.5], \theta_0^* = [0.5, -1, 1, 0.5]$$

```
t_vals = linspace(0, 1, 100)
```



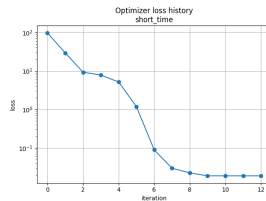
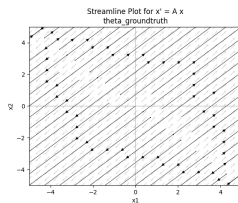
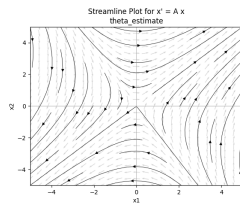
```
final loss = 0.0186
```

```
final estimate = [-0.49, 0.99, -1.00, -0.49]
```

ill-conditioned

$$\theta_{\text{groundtruth}} = [1, 1, 1, 1], \theta_0^* = [1, 1, 0, 1]$$

`t_vals = linspace(0, 1, 100)`



`final loss = 0.0188`

`final estimate = [ 0.82, 1.17, 2.02, -0.028]`

Questions?